

Elizabeth Dietrich

University of California, Berkeley

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Position

University of California, Berkeley

GRADUATE STUDENT RESEARCHER

Electrical Engineering and Computer Sciences

Berkeley, USA

08/2023 – Today

Interests

Methods

reachability analysis · formal methods · data-driven control

Topics

autonomous systems · control theory · statistical learning

Education

University of California, Berkeley

PHD IN ELECTRICAL ENGINEERING AND COMPUTER SCIENCES

Advisor: Murat Arcak

Berkeley, CA, USA

08/2023 – Today

Indiana University, Bloomington

B.S. IN COMPUTER SCIENCE, MINOR IN MATHEMATICS

Thesis: A beginner's guide to Iris, Coq and separation logic, Advisor: Dennis Shasha

Bloomington, IN, USA

08/2017 – 05/2021

Visits

Norwegian University of Science and Technology (NTNU)

VISITING RESEARCHER

MC Lab (Asgeir J Sørensen and Roger Skjetne)

Trondheim, Norway

06/2025 – 05/2026

Norwegian University of Science and Technology (NTNU)

VISITING RESEARCHER

MC Lab (Asgeir J Sørensen)

Trondheim, Norway

05/2024 – 07/2024

Publications

* first author(s)

PREPRINTS

[P3] **Elizabeth Dietrich***, Liam M. Imagawa*, Hanna Krasowski, Aurora Haraldsen, Murat Arcak, and Kristin Y. Petterson. *Expressiveness, Equivalence, and Uncertainty in Velocity Obstacles and Closest Point of Approach Metrics*. arXiv:2610.07371. 2026

[P2] **Elizabeth Dietrich***, Hanna Krasowski*, Emir Cem Gezer, Roger Skjetne, Asgeir Johan Sørensen, and Murat Arcak. *Staying on Spec: Real-Time Monitoring under Uncertainty with a Maritime Case Study*. arXiv:2608.02811. 2026

[P1] Varun Madabushi*, **Elizabeth Dietrich***, Hanna Krasowski, and Maegan Tucker. *Finite-Step Invariant Sets for Hybrid Systems with Probabilistic Guarantees*. arXiv:2604.05102. 2026

JOURNAL ARTICLES

[J1] Hanna Krasowski*, **Elizabeth Dietrich***, Emir Cem Gezer, Roger Skjetne, Asgeir Johan Sørensen, and Murat Arcak. "pac-STL: PAC-Bounded Signal Temporal Logic from Data-Driven Reachability Analysis". In: *IEEE Control Systems Letters* 10 (2026), pp. 2191–2196. doi: 10.1109/LCSYS.2026.3715556

CONFERENCE PAPERS

- [C5] **Elizabeth Dietrich***, Hanna Krasowski*, and Murat Arcak. “Probably Approximately Correct (PAC) Guarantees for Data-Driven Reachability Analysis: A Theoretical and Empirical Comparison”. In: *Accepted for Proc. of the IEEE Conference on Decision and Control (CDC)*. 2026
- [C4] **Elizabeth Dietrich***, Hanna Krasowski, Vegard Flovik, and Murat Arcak. “Importance Sampling for Statistical Certification of Viable Initial Sets”. In: *Accepted for Proc. of the IEEE Conference on Decision and Control (CDC)*. 2026
- [C3] **Elizabeth Dietrich***, Rosalyn Devonport, Stephen Tu, and Murat Arcak. “Data-Driven Reachability with Scenario Optimization and the Holdout Method”. In: *2025 IEEE 64th Conference on Decision and Control (CDC)*. 2025, pp. 3925–3931. doi: 10.1109/CDC57313.2025.11312065
- [C2] **Elizabeth Dietrich***, Emir Cem Gezer*, Bingzhuo Zhong, Murat Arcak, Majid Zamani, Roger Skjetne, and Asgeir Johan Sørensen. “Symbolic Control for Autonomous Docking of Marine Surface Vessels”. In: *2025 IEEE 64th Conference on Decision and Control (CDC)*. 2025, pp. 3021–3026. doi: 10.1109/CDC57313.2025.11312758
- [C1] **Elizabeth Dietrich***, Alex Devonport, and Murat Arcak. “Nonconvex scenario optimization for data-driven reachability”. In: *Proceedings of the 6th Annual Learning for Dynamics Control Conference*. Vol. 242. Proceedings of Machine Learning Research. PMLR, 15–17 Jul 2024, pp. 514–527. <https://proceedings.mlr.press/v242/dietrich24a.html>

Invited Talks

- 2025** NSF Frontier Review, UC Berkeley · SFI Autoship Review, NTNU · Marine Cybernetics Seminar, NTNU · ANSR Seminar, UC Berkeley, online
- 2022** Informatics, Computing, and Engineering Seminar, IUB · Intelligent Systems Symposium, JHU APL

Mentoring

STUDENT PROJECTS AND THESES

- | | | |
|-----------|---|------------|
| 2026- | Liam Imagawa (B.S.) | UCB |
| 2025-2026 | Petter Dalhaug (M.S.) | NTNU & UCB |
| 2025-2026 | Athul Krishnan (B.S.) & Chloe Yejin Lynn (B.S.) | UCB |
| | NSF IRES Project on Autonomous Marine Operations for the Arctic | |

Professional Experience

- 09/2026 – **AI Research Intern** · DNV (Det Norske Veritas)
- 08/2021 – 07/2023 **Associate Staff: Research Scientist** · Johns Hopkins University Applied Physics Laboratory
- 05/2020 – 08/2021 **Undergraduate Research Assistant** · New York University, Courant Institute
- 05/2019 – 08/2019 **Software Engineer Intern** · Amazon

Teaching Experience

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|------------------|---|
| Fall 2026 | Introduction to Embedded and Cyber Physical Systems (EECS 149) , Graduate Teaching Assistant |
| Spring 2025 | Optimization of Engineering Systems (MAS-E 238C) , Graduate Teaching Assistant (online) |
| Fall 2024 | Analysis and Control of Nonlinear Systems (MAS-E 250A) , Graduate Teaching Assistant (online) |
| Spring 2022 | MATLAB II , STEM Academy Instructor JHUAPL |
| Fall 2020 | Training for Paraprofessionals in Student Affairs (EDUC-U208) , Undergraduate Teaching Assistant |
| Fall 2018 & 2019 | Introduction to Computer Science (CSCI-C211) , Undergraduate Teaching Assistant |

Service to the Profession

PEER REVIEW

Conferences (reviewed 5 submissions): Conference on Decision and Control (CDC), International Conference on Robotics and Automation (ICRA)

Journals (reviewed 7 submissions): Control System Letters (L-CSS), Nonlinear Analysis: Hybrid Systems (NAHS), Transactions on Automation Science and Engineering (T-ASE), Transactions on Automatic Control (TAC), Transactions on Control Systems Technology (TCST)

COMMUNITY BUILDING

2024 -	EECS Equal Access to Application Assistance (EEAAA) , Assist underrepresented prospective PhD students on their graduate school applications to help equalize access to guidance on the application process.	UCB
2026	EECS PhD Visit Days Coordinator , Coordinate visit days for close to 200 students admitted to the EECS PhD program.	UCB
2026	Application reviewer for Berkeley EECS MEng program , Read and evaluate applications for admission to Berkeley's EECS professional master programs.	UCB
2025	EECS PhD CIR & CPSDA Visit Days , Organization of activities for prospective PhD students in controls, automation, learning, and embedded systems.	UCB

Awards, Fellowships, & Grants _____

- **2026 Graduate Diversity & Community Fellowship**, *University of California, Berkeley*.
Awarded by the Office for Graduate Diversity to advance diversity and inclusion initiatives by providing peer mentorship, leading prospective student outreach, and coordinating events.
- **2026 D-Lab Data Science & AI Fellowship**, *University of California, Berkeley*.
Awarded to graduate students to design and lead technical workshops that advance data science and AI literacy across the UC Berkeley campus.
- **2026 EECS Departmental Chair's Graduate Award**, *University of California, Berkeley*.
- **2025 Fernström Fellowship**, *University of California, Berkeley*.
Awarded to support graduate research in the Nordic countries.
- **2023 NSF Graduate Research Fellowship (GRFP)**, *National Science Foundation*.
- **2021 Kate Hevner Mueller Senior Award**, *Indiana University, Bloomington*.
Awarded to undergraduates who demonstrated exceptional leadership and made a significant impact on the Indiana University community.
- **2020 Goldwater Scholar**, *Barry Goldwater Scholarship and Excellence in Education Foundation*.